ADVANCED ROBOTIC PERCEPTION

1: INTRO, CONVOLUTION, AND EDGES

RIKKE GADE 2026

AALBORG UNIVERSITY
DENMARK

Welcome – practical information

Who are you?

Who am I?

- Associate professor, PhD in Computer Vision
- Office: Rendsburggade 14, 6.311 (CREATE / City Campus)
- Email: rg@create.aau.dk

Other people:

- Stefan Hein Bengtson, PostDoc. Teaching two lectures.
- Kasper Fuglsang Jordt, research assistant. Teaching assistant – helping with exercises.
  - kfjo@create.aau.dk



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Practical information

Logistics:

- Lectures at FRB
  - Many lectures will be shared with students from Computer Engineering (AI, Vision, and Sound specialisation)
- Exercises in group rooms
  - The TA will be around and visit you regularly
  - If you don’t see him, just write an email or Teams message to me to call for help
- Course information on Moodle (will be updated throughout the semester)

Practical information

Lecture exercises

- Most lectures have hands-on coding exercises related to the topic of the lecture
- Later we will also work on mini-projects in groups
  - Will be introduced later



Practical information

A note on literature:
- In the lecture I give an introduction to the topic and discuss the context, and explain some methods
- Not necessary to read the listed literature before a lecture
- You will need the literature when slides are insufficient or when you need to implement the methods
  - The book by Richard Szeliski has many references you may follow to learn more
  - When introducing an algorithm, I will refer to the original paper

Tools:
- OpenCV library for C++ or Python
- (MATLAB)

Exam:
- Individual oral exam
- January 2026
- Assessment: 7-point scale

Practical information

Video lectures:

- Some lectures may be given as videos instead of in-person lectures
- We will still be around in the second half of the lecture time to help with exercises and to answer questions
- Please check the instructions on Moodle under each individual lecture

Advanced Robotic Perception

So what is this all about?

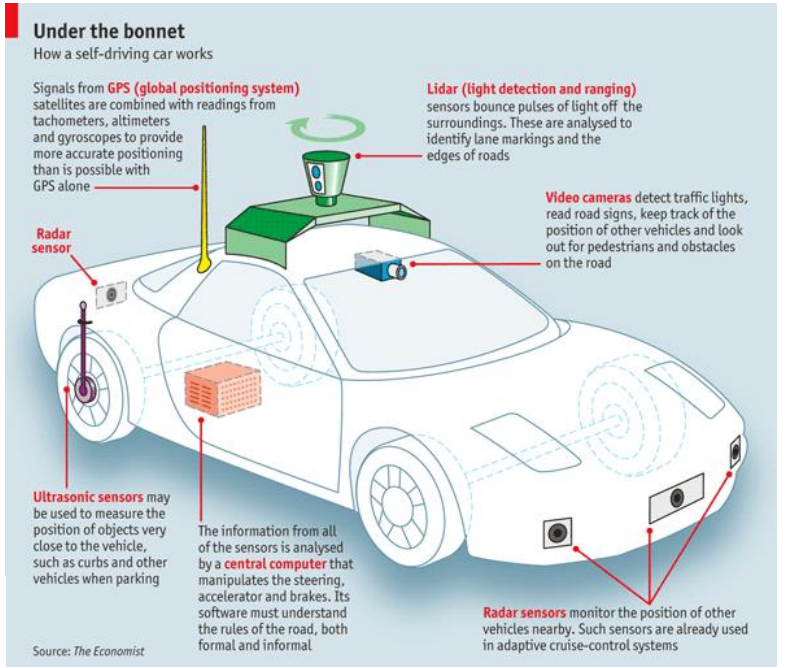
Do we need this to build robots?

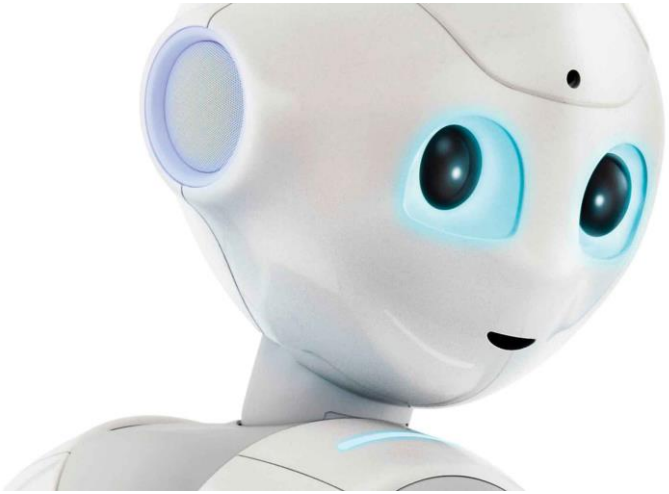


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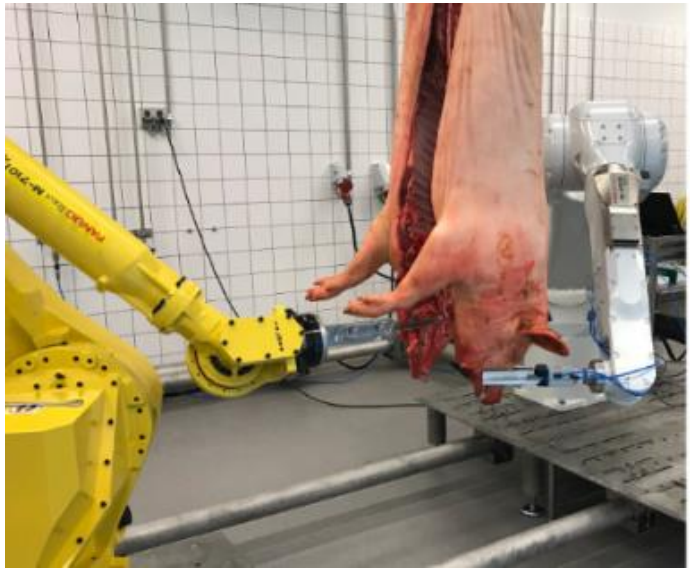












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Learning objectives

Knowledge

- Must be able to explain the principles behind robust feature point algorithms
- Must have knowledge of feature selection and reduction methods
- Must have knowledge of motion analysis principles
- Must be able to understand tracking frameworks
- Must be able to understand how advanced perception is integrated into robotic systems (e.g visual servoing, obstacle avoidance)

Skills

- Must be able to apply sliding window approaches based on advanced features to detect objects
- Must be able to apply stereo vision to generate 3D data from two or more cameras
- Must be able to apply model-based approaches to estimate the 3D pose of objects and people
- Must be able to apply pattern recognition methods to classify object types and activities
- Must be able to integrate advanced perception into robotic systems

Competences

- Must be able to analyse a specific problem within mobile robotics and based upon this select, implement and evaluate an appropriate computer vision approach

Course content

Lectures:

1: Intro + convolutions and edges
2: Robust features, selection and reduction
3: Machine learning for vision systems
4: Convolutional Neural Networks
5: Transformers and other advanced networks
6: Object detection and segmentation
7: Motion analysis
8: Tracking
9: Pose estimation and point clouds
10: Depth sensing
11: Camera-robot calibration

Mini-projects

Course content

Starting point:

- (ROB3 Robotic Perception)
- Knowledge on basic image processing
- Knowledge on image acquisition and sensor technologies

Images can be obtained from various sensor types

- E.g., RGB, monochrome, NIR, thermal, depth, linescan…
- 3-channel vs. 1-channel
- 2D image vs. point cloud

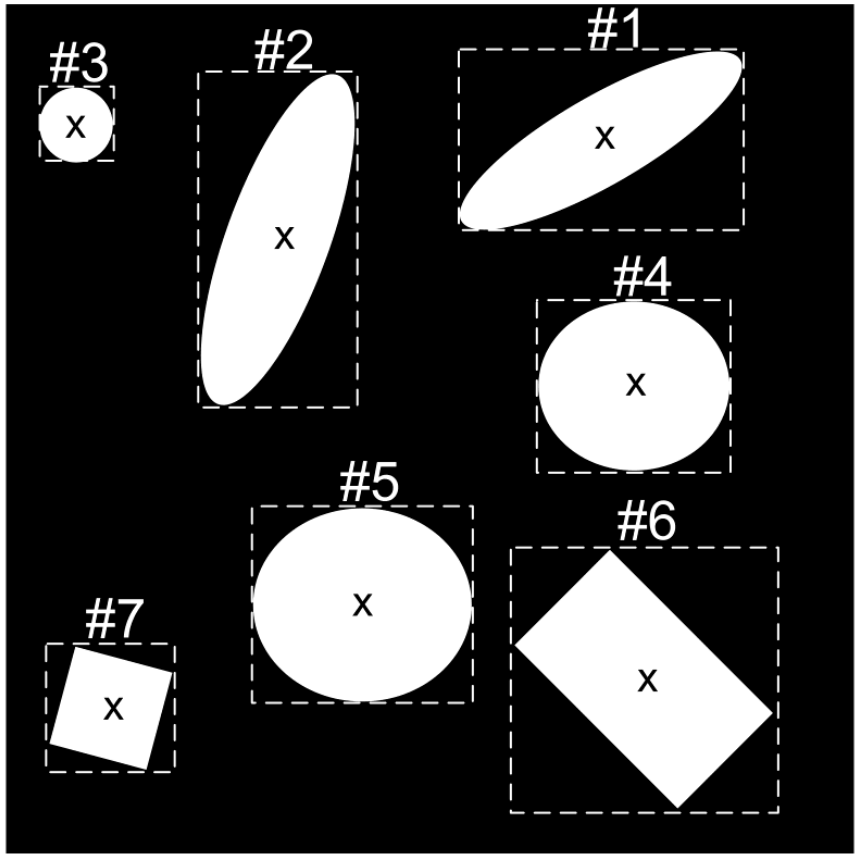
Agenda today

- Short recap on features and image correlation
- Edge filters
- Canny edge detector



Features

- Classic features for classification:
  - Area
  - Height/width ratio
  - Circularity
  - Holes
  - Perimeter
  - Bounding box
  - …
- Describe properties of an extracted BLOB

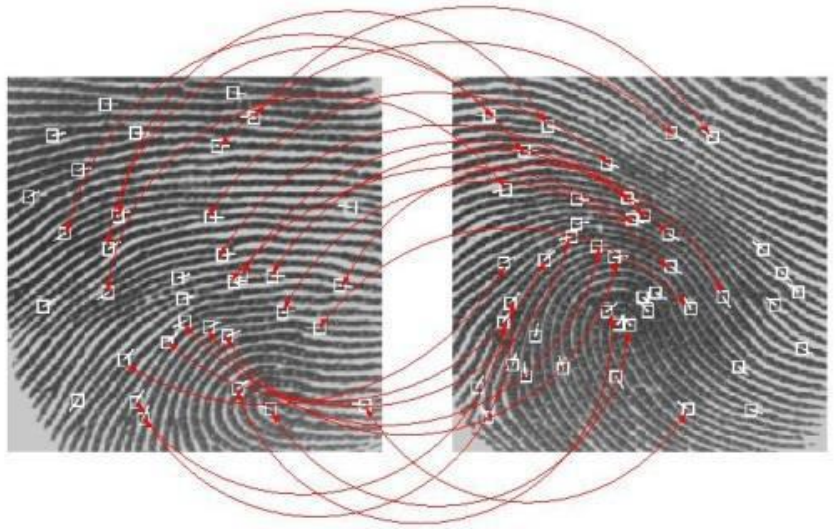


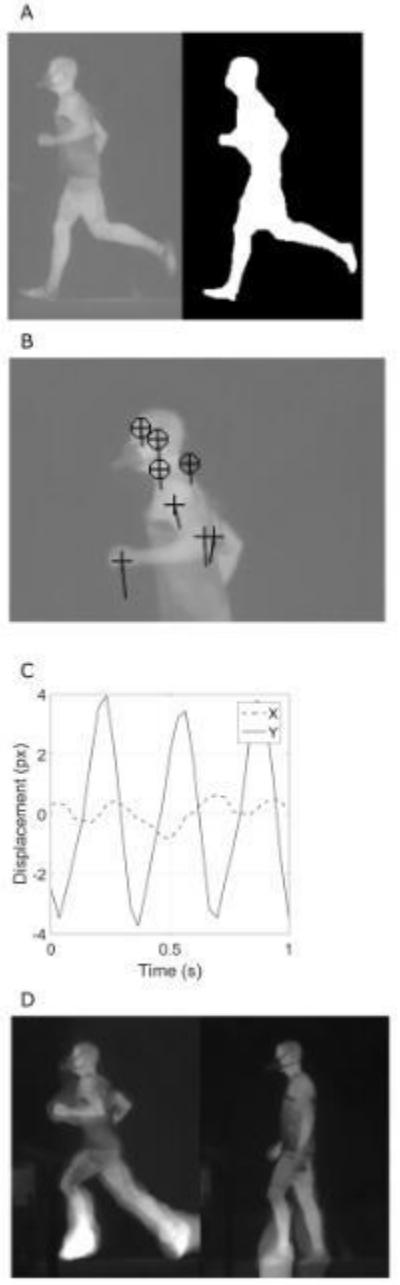
Features - applications

Features – a way to describe areas or objects in an image

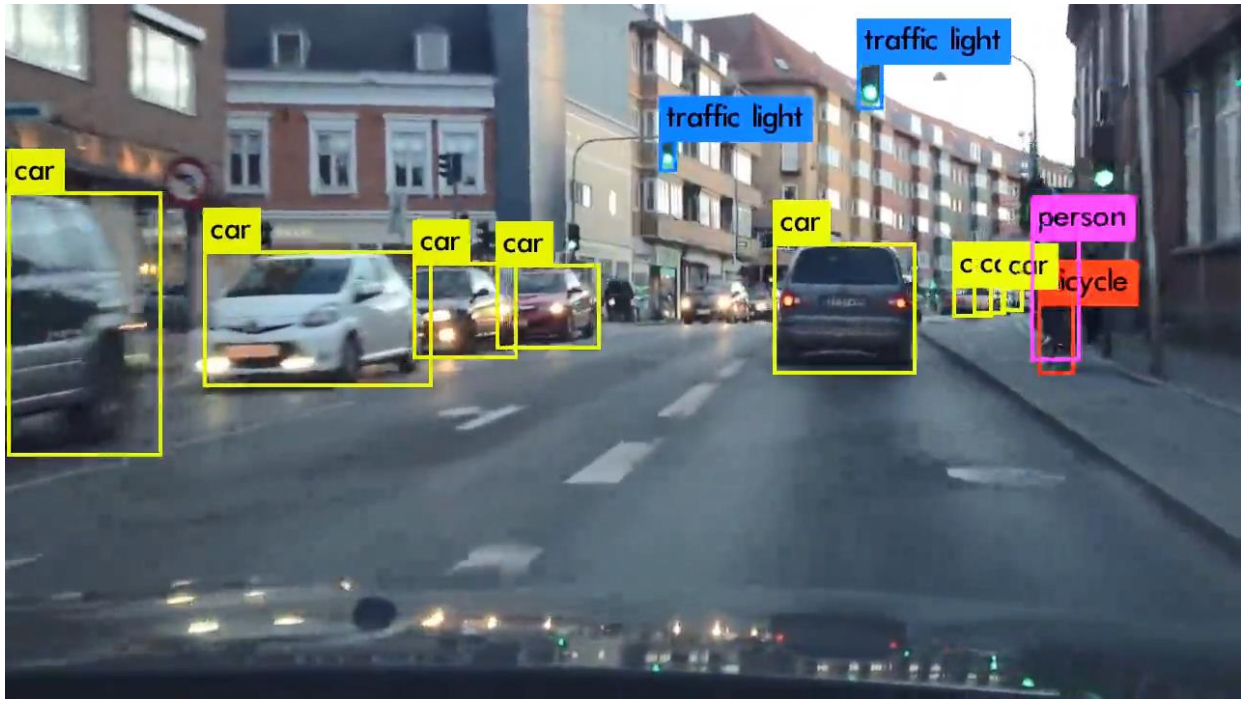
Used for, e.g.:

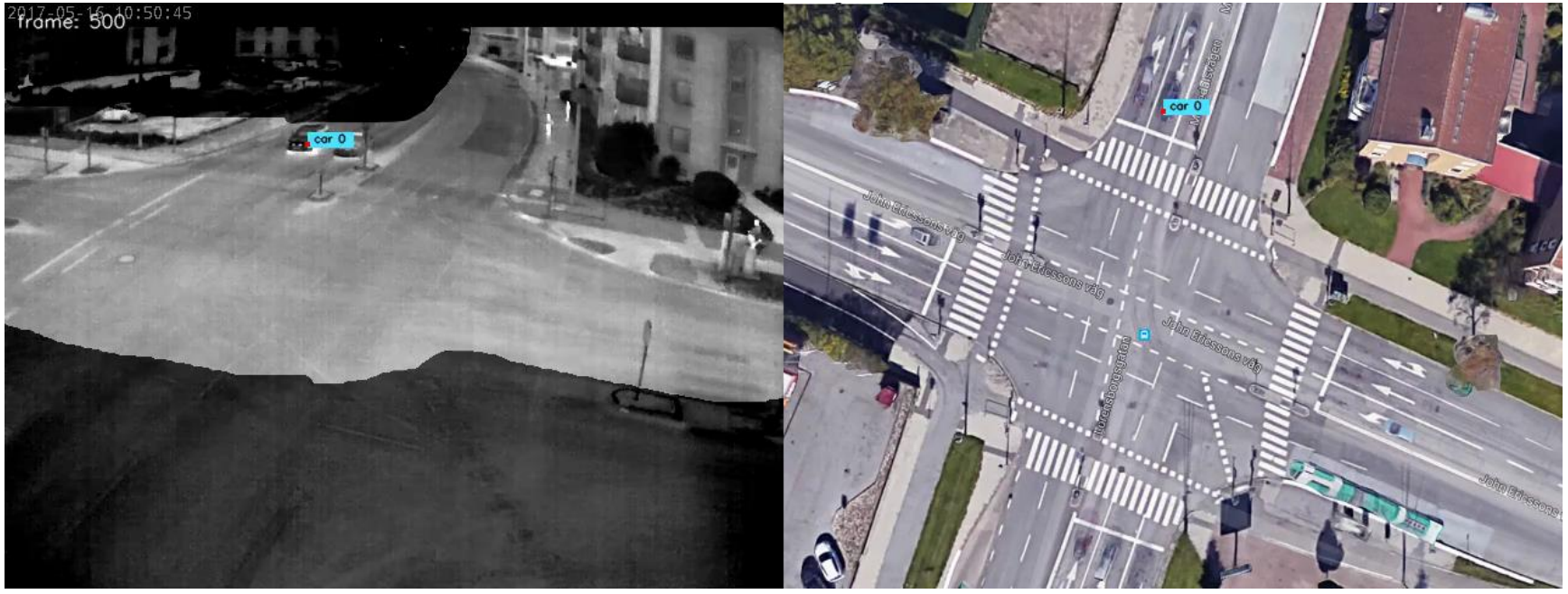
- Detection/recognition
- Stitching/matching/aligning images
- Tracking/motion analysis
- Model construction
- Stereo vision
- …





Features - applications





Features - applications

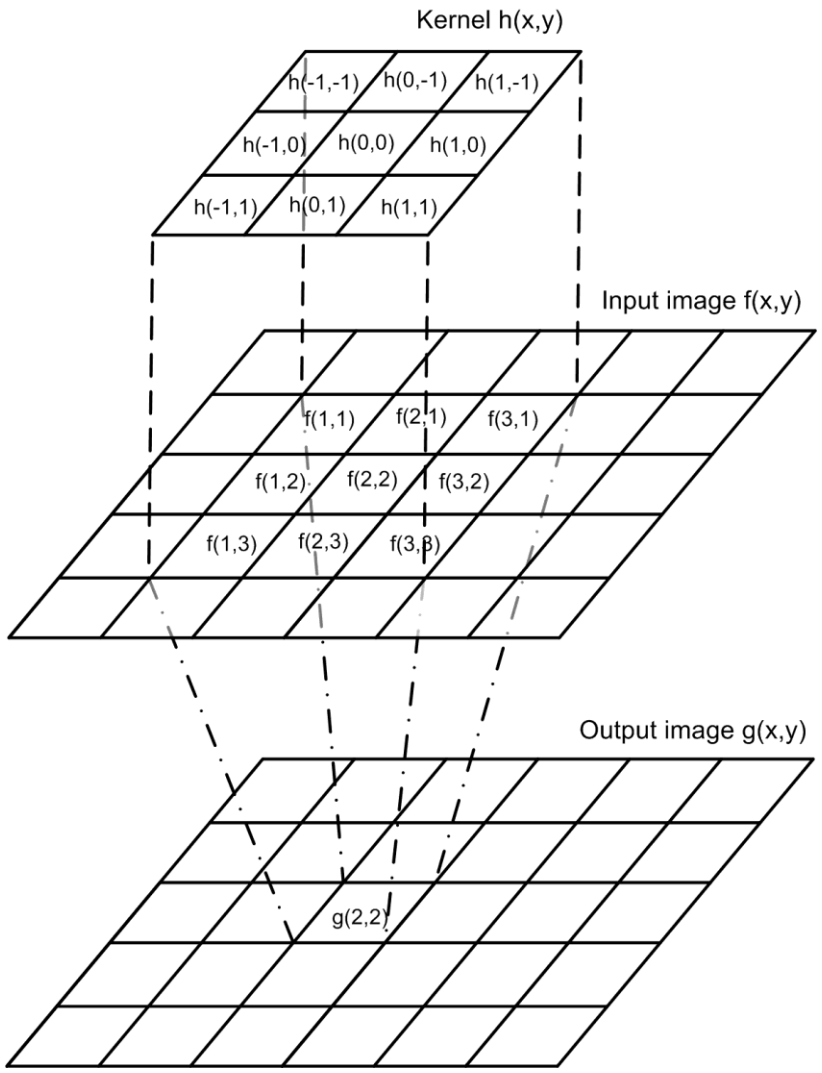


Image correlation

Use a kernel to filter an image

$$g(x,y) = h \circ f(x,y) = \sum_{-n}^{n} \sum_{-m}^{m} h(m,n) f(x+m, y+n)$$

g(x,y): output, h: filter, ∘: correlation, f(x,y): input, n: radius (height) (don't count the centre pixel), m: radius (width)

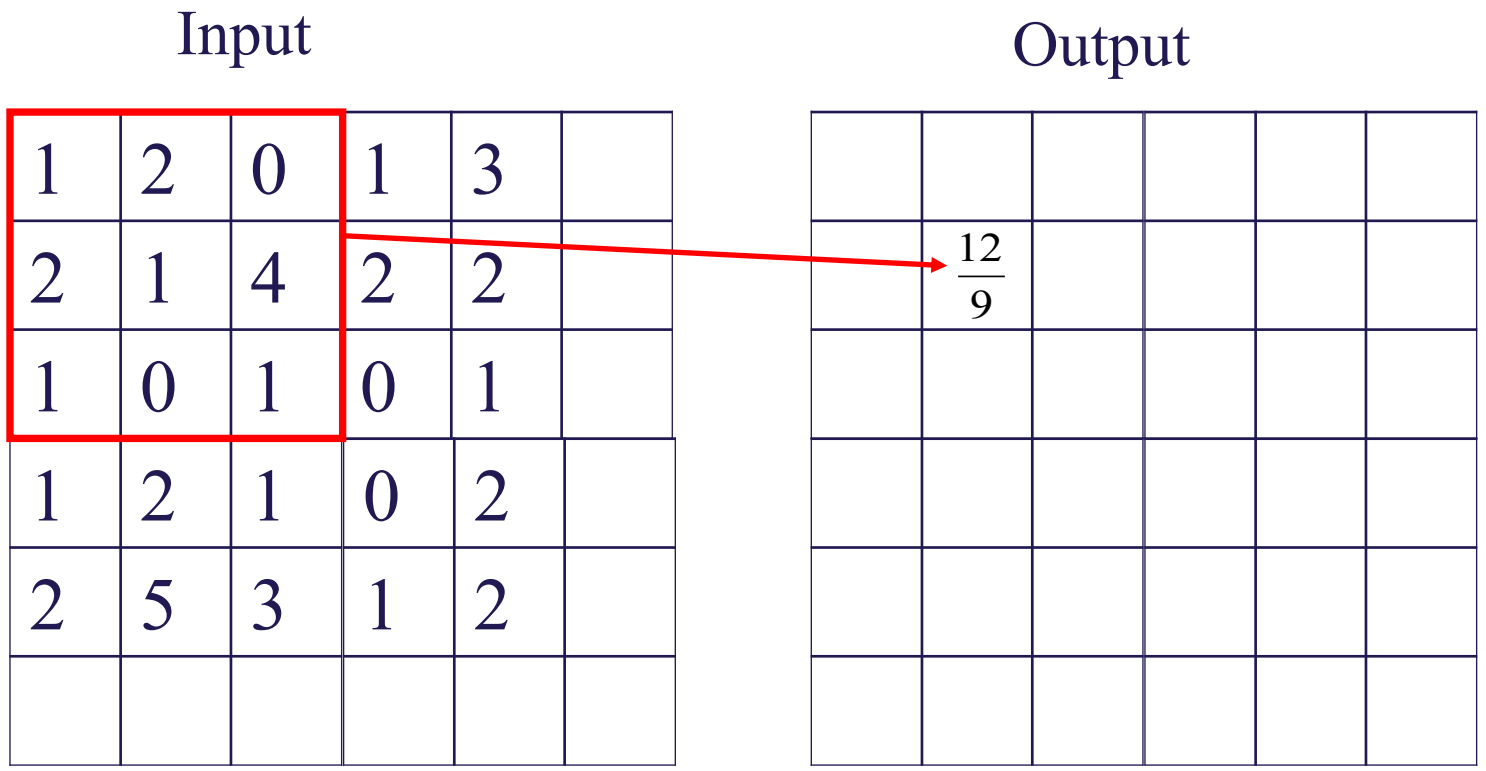


Examples of application:

- Blurring/smoothing
- Edge detection
- Template matching

Image correlation

- Simple mean filter example



$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Image correlation

- Simple mean filter example

Input

1	2	0	1	3	
2	1	4	2	2	
1	0	1	0	1	
1	2	1	0	2	
2	5	3	1	2	

Output

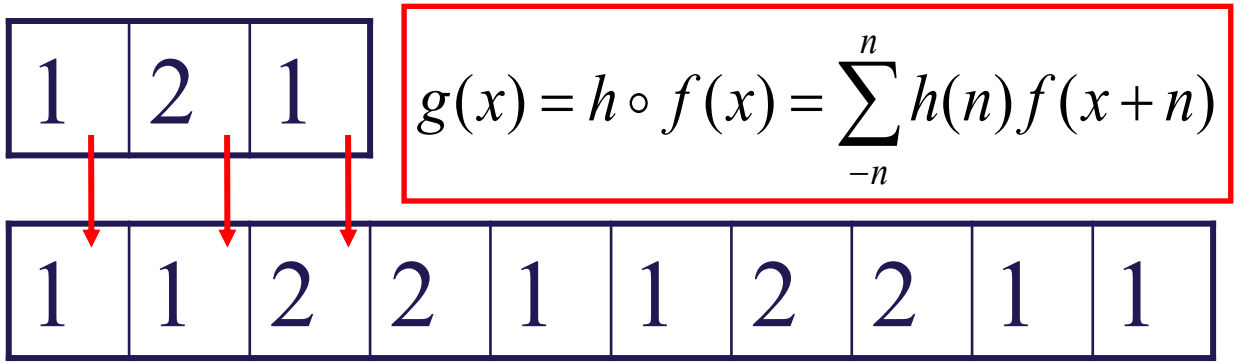
	$\frac{12}{9}$	$\frac{11}{9}$			

$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

Convolution vs. correlation

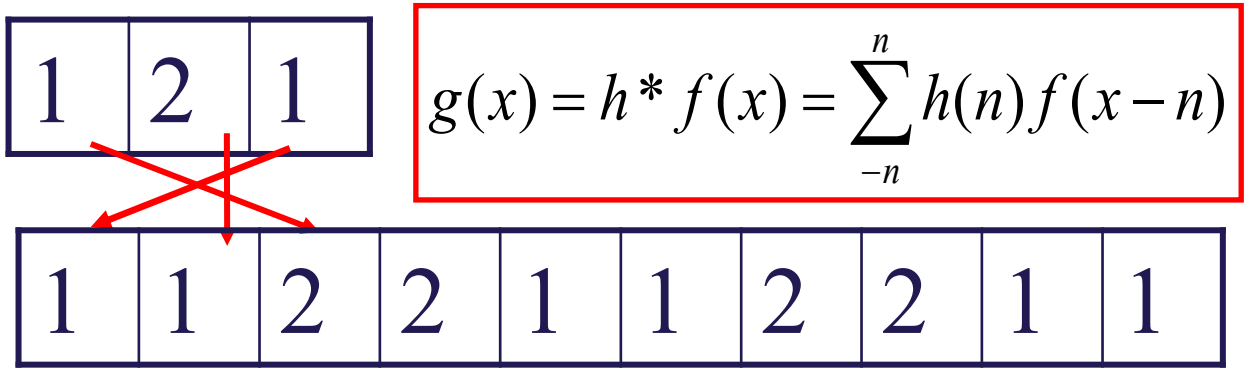
- In image processing we use CORRELATION but often call it CONVOLUTION!
- Note: When the filter is symmetric: correlation = convolution

Correlation



$$g(x) = h \circ f(x) = \sum_{-n}^{n} h(n) f(x+n)$$

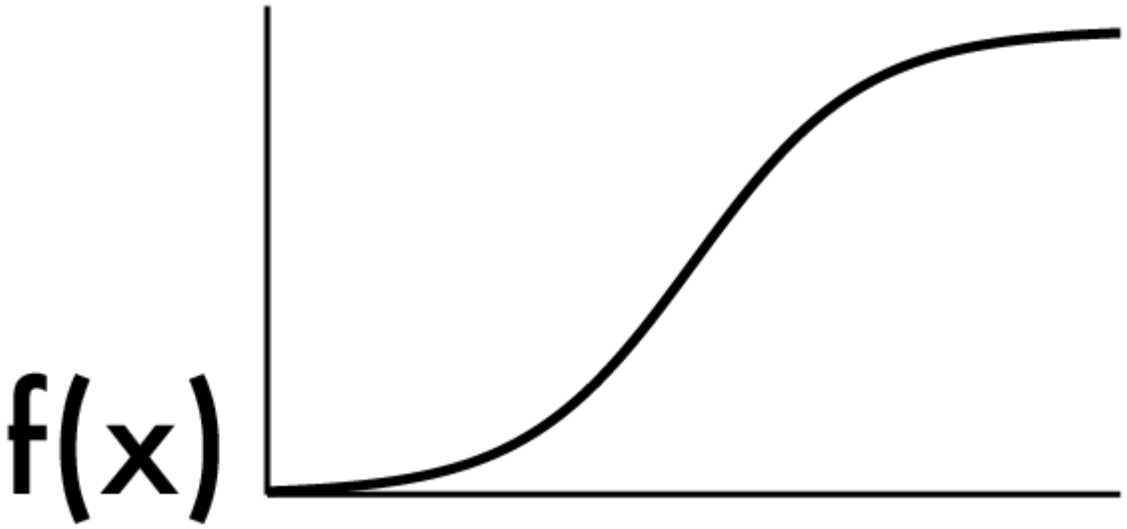
Convolution



$$g(x) = h * f(x) = \sum_{-n}^{n} h(n) f(x-n)$$

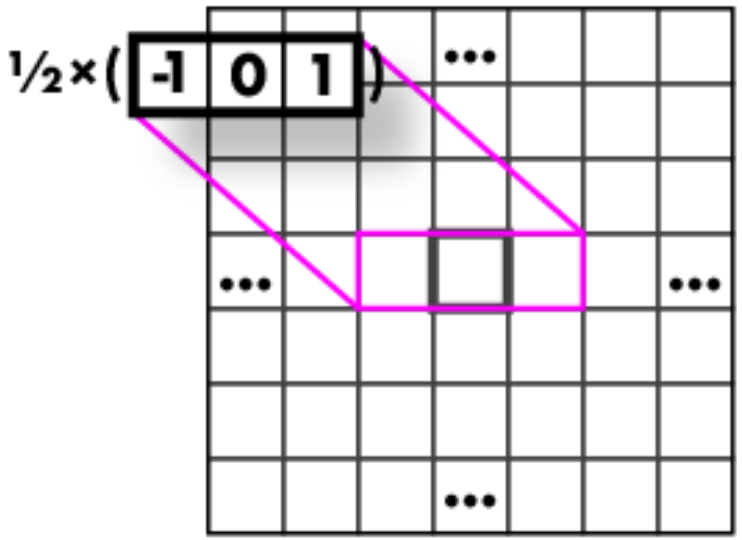
Edge detection

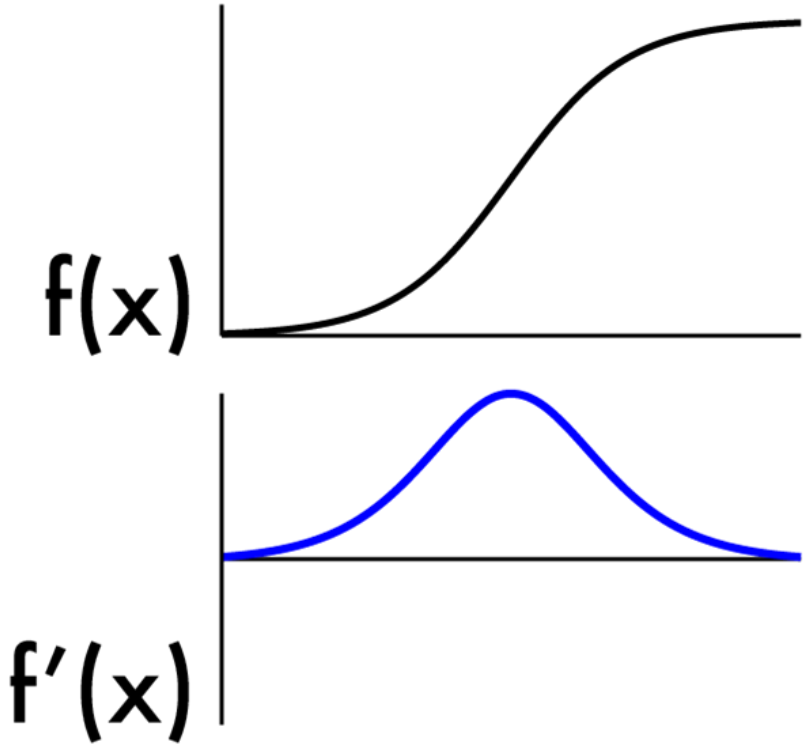
- Image is a function
- Edges are rapid changes in this function



Edge detection

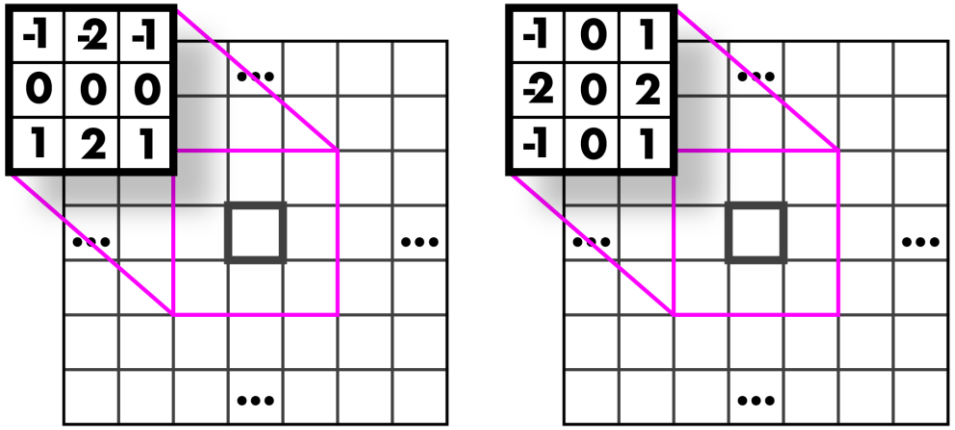
- Could take derivative
- Edges = high response

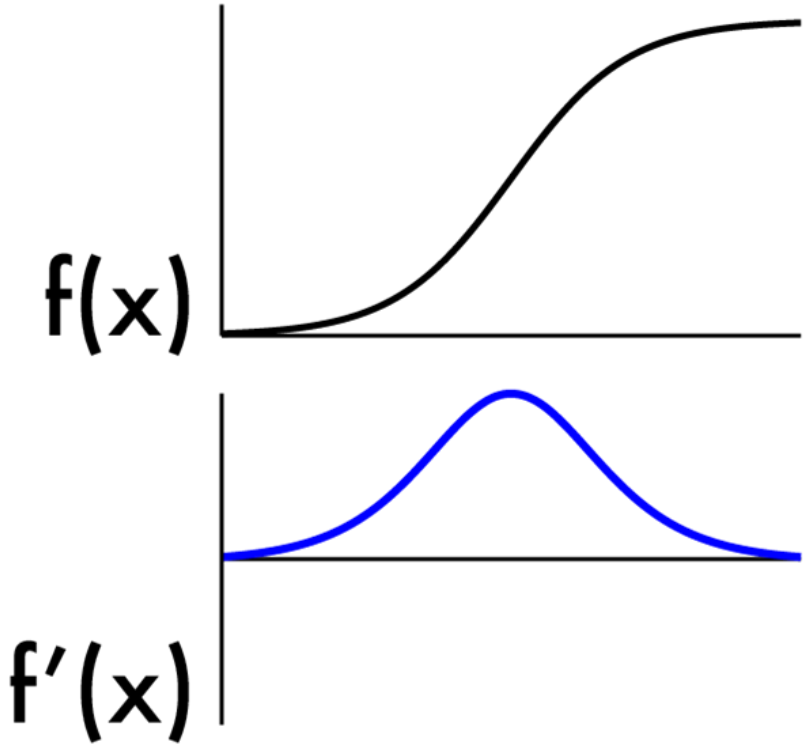




Edge detection

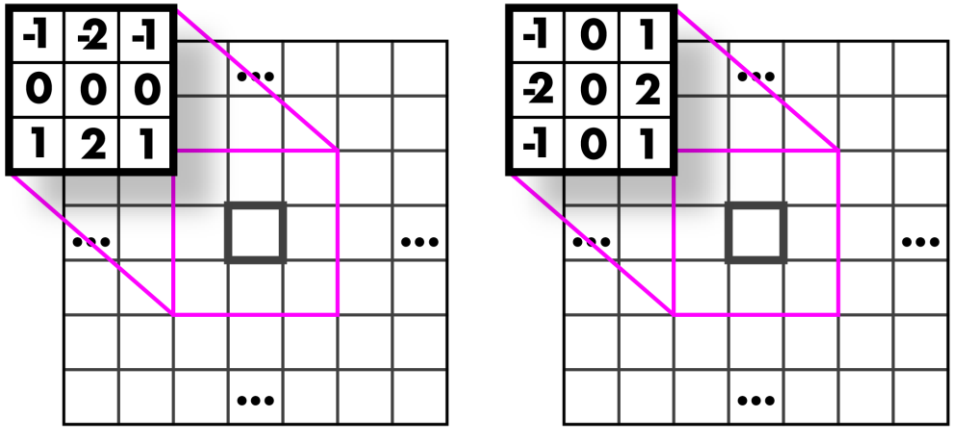
- Sobel filters
  - Gradient + smoothing
- What is an edge?
  - High response

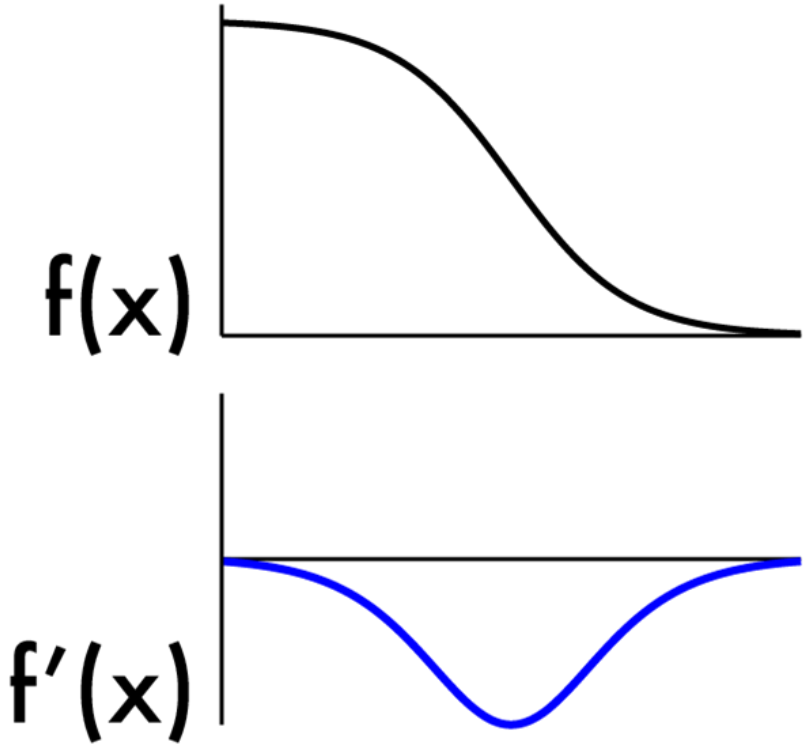




Edge detection

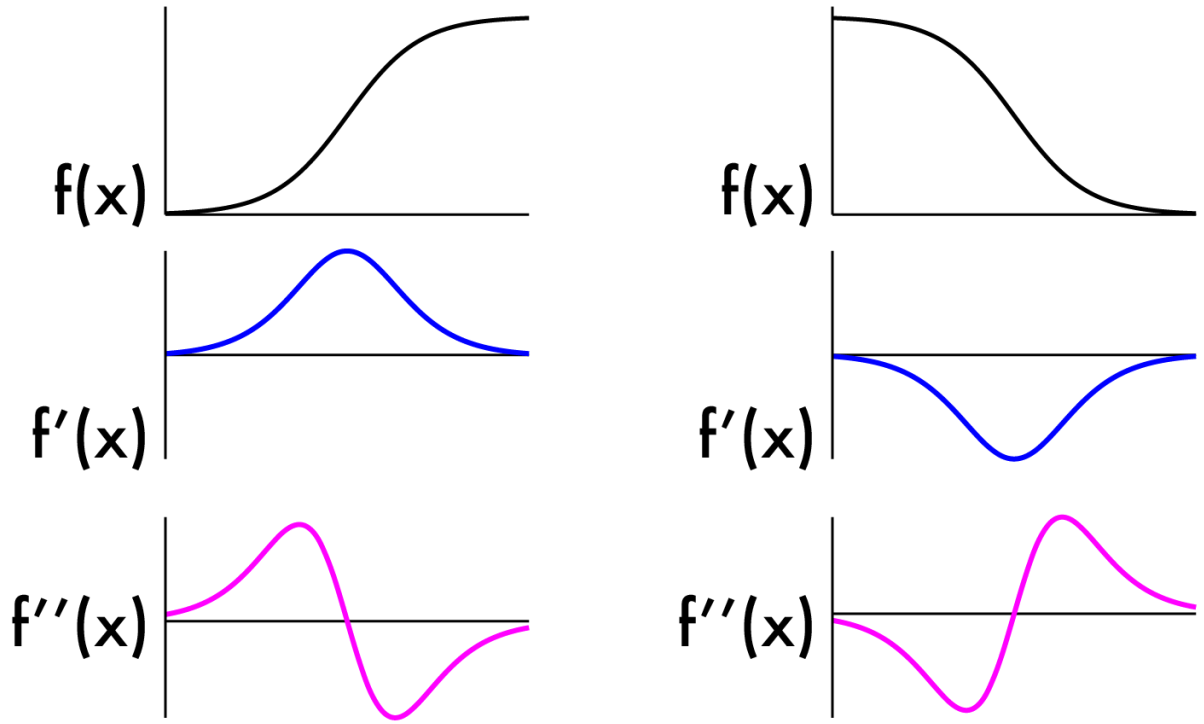
- Sobel filters
- What is an edge?
  - ~~High response~~
  - Extrema





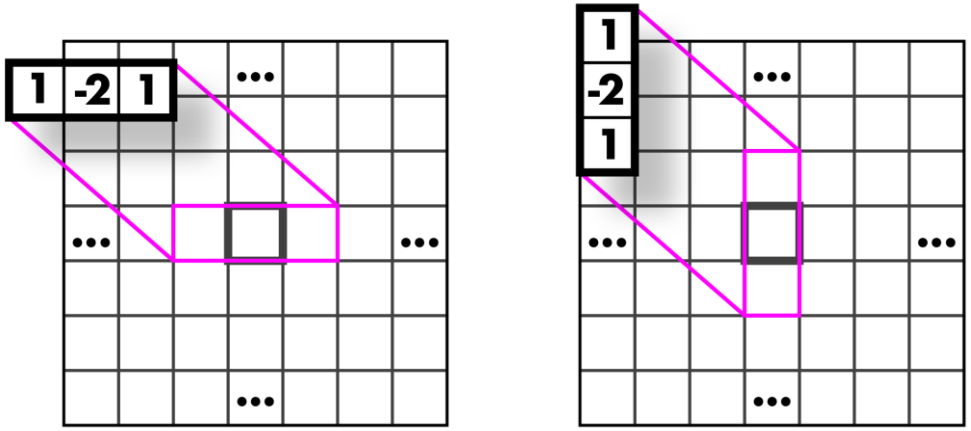
Edge detection

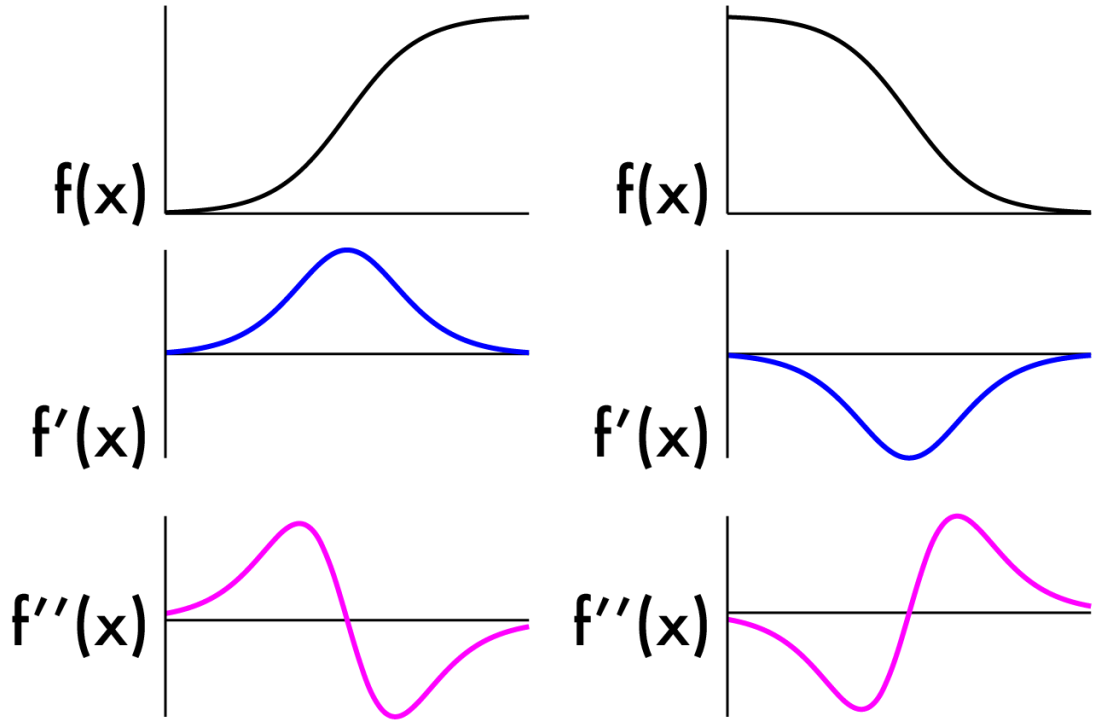
- 2nd derivative: Crosses zero at extrema



Edge detection

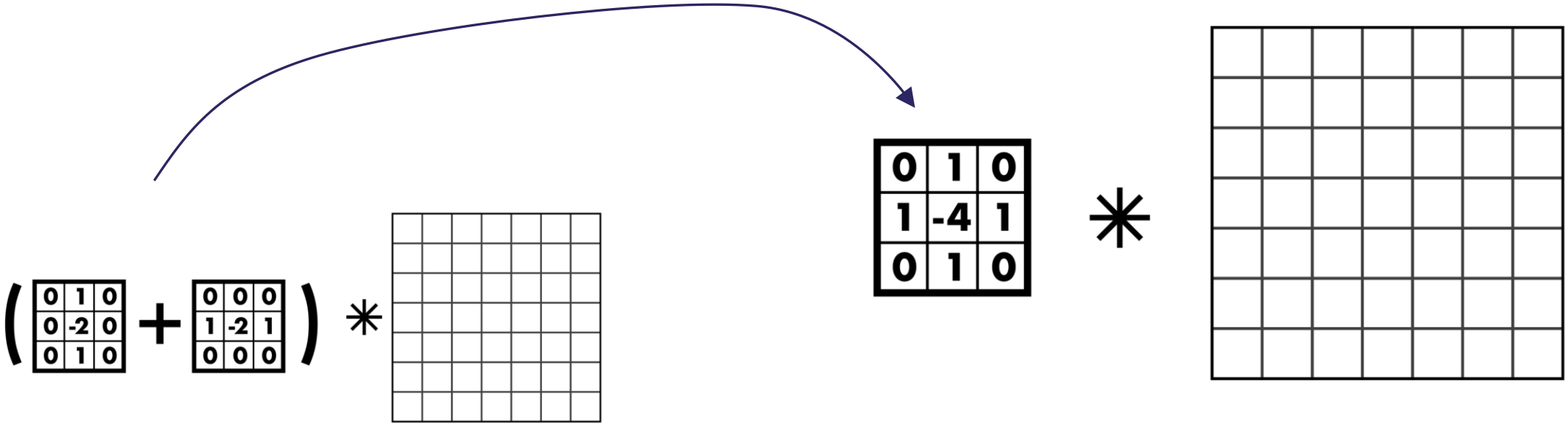
- Laplacian: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$
- Estimated 2nd derivative:





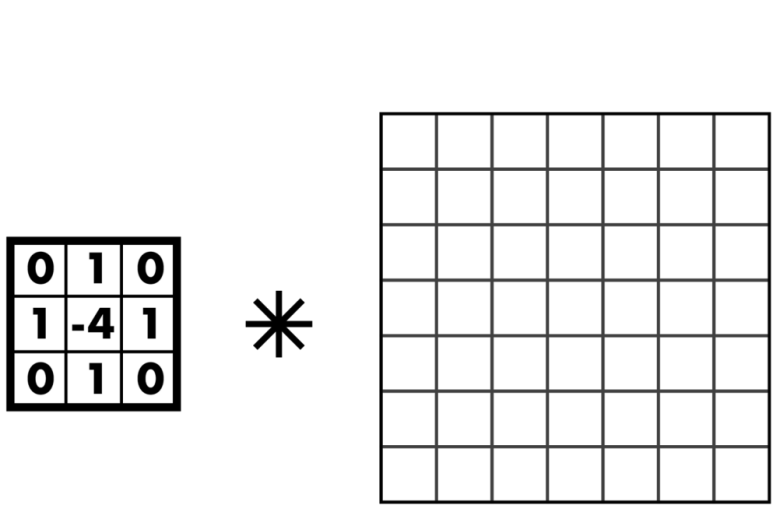
Edge detection

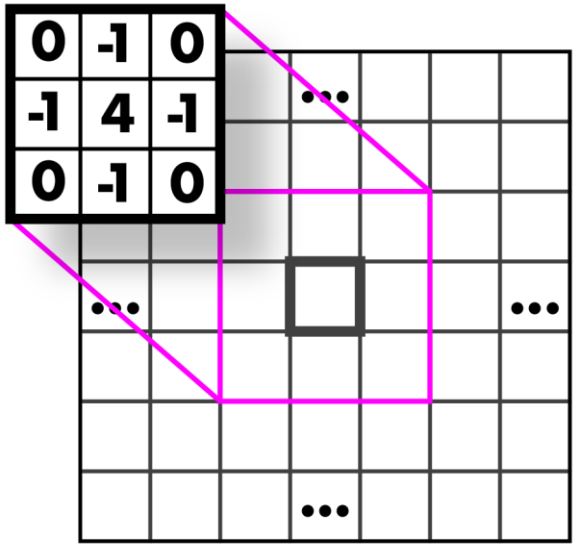
- Laplacian: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$



Edge detection

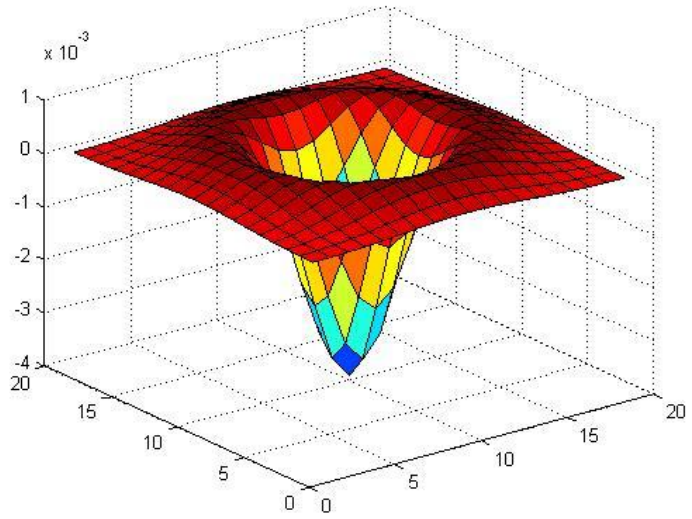
- Laplacian: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$
- Negative vs. positive Laplacian





Edge detection

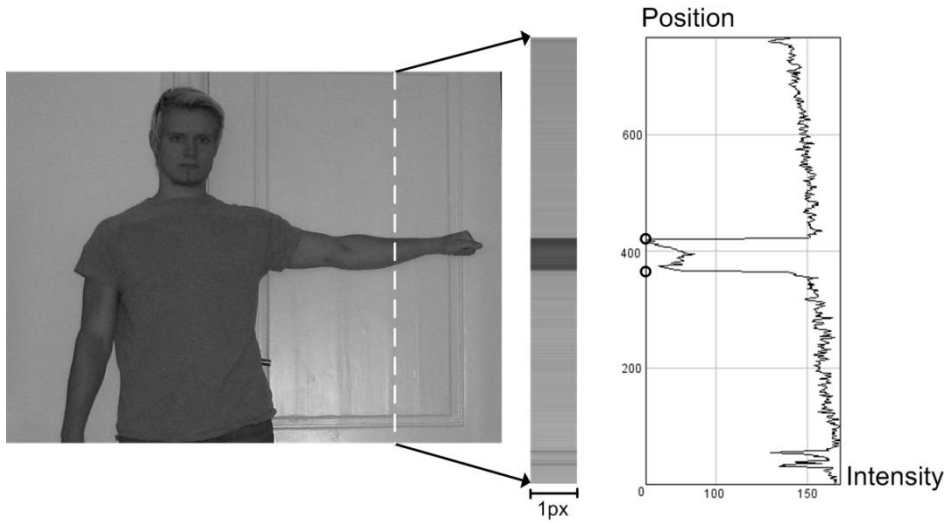
- 2nd derivative is very sensitive to noise!
- Use Gaussian smoothing
- Combined: Laplacian of Gaussian, LoG

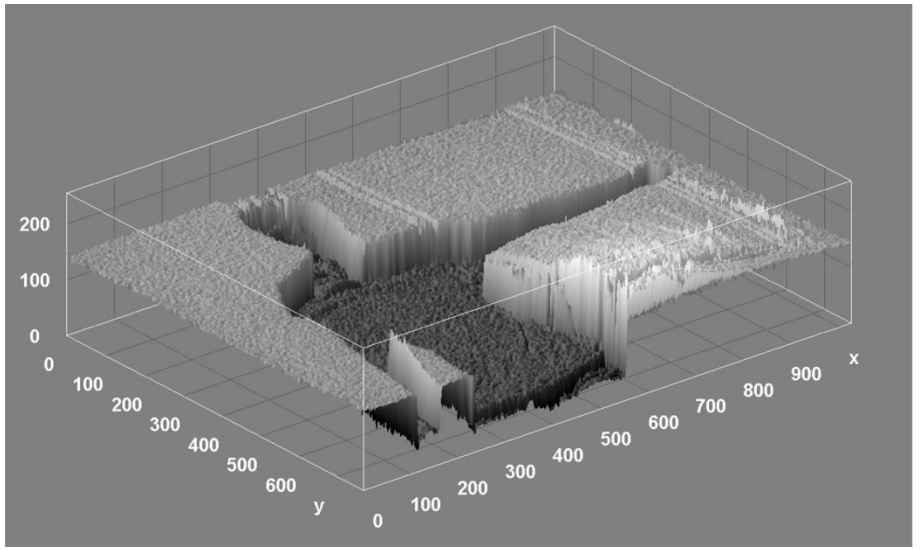


Edge detection

Another approach:

- Image is a function with high and low frequency components
  - Think in terms of Fourier transform
- Edges are high frequency changes
- Maybe we want to find edges of a specific size
  - I.e., specific frequency





Difference of Gaussian (DoG)

Gaussian function = low pass filter

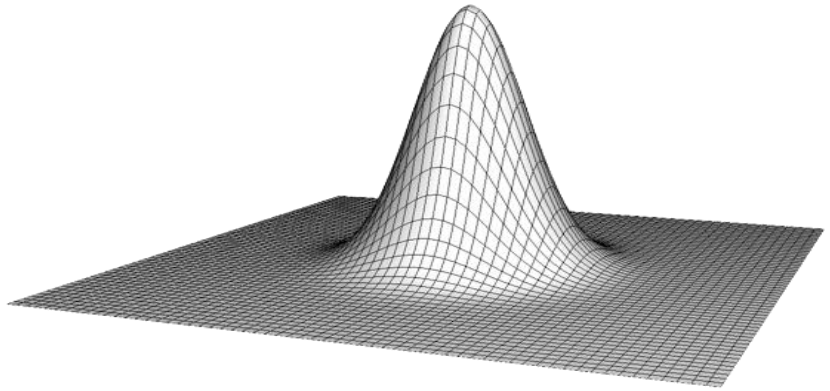
- Strongly reduces components with frequency f > σ

$g(\sigma) * I \rightarrow$ low frequency components

$I - (g(\sigma) * I) \rightarrow$ high frequency components

$g(\sigma_1) * I - g(\sigma_2) * I \rightarrow$ components between these frequencies

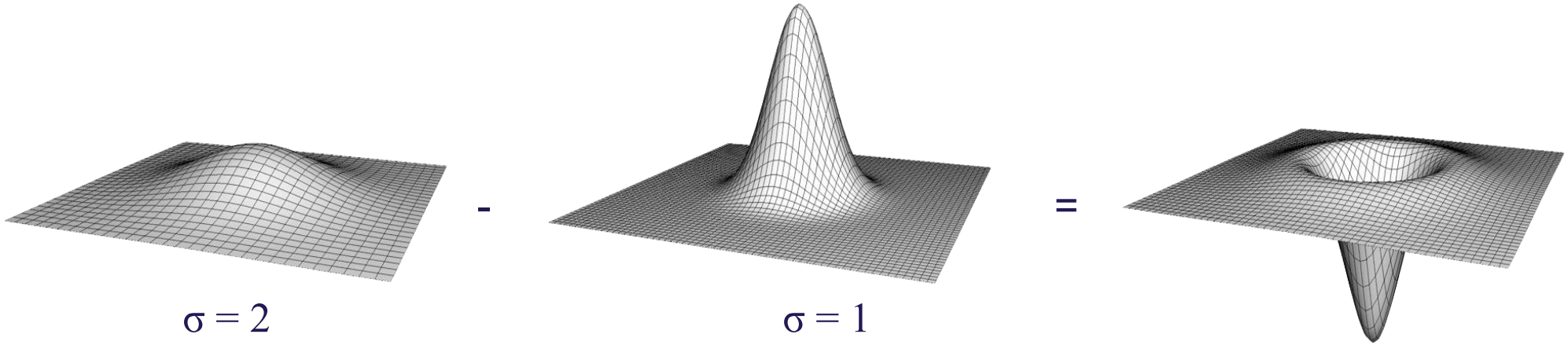
$$g(\sigma_1) * I - g(\sigma_2) * I = [g(\sigma_1) - g(\sigma_2)] * I$$



Gaussian function $g(\sigma)$

Difference of Gaussian (DoG)

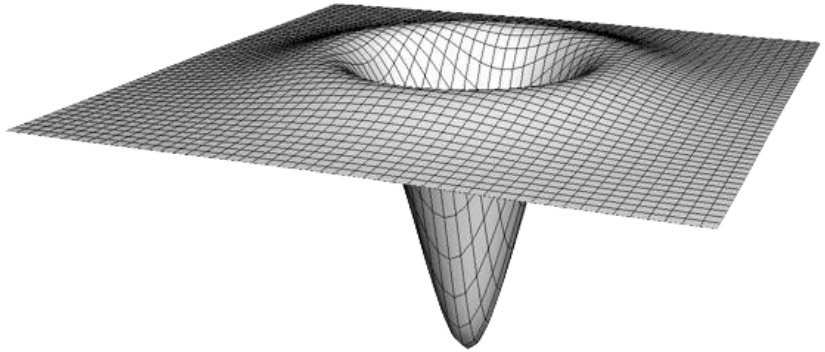
$$g(\sigma_1) * I - g(\sigma_2) * I = [g(\sigma_1) - g(\sigma_2)] * I$$

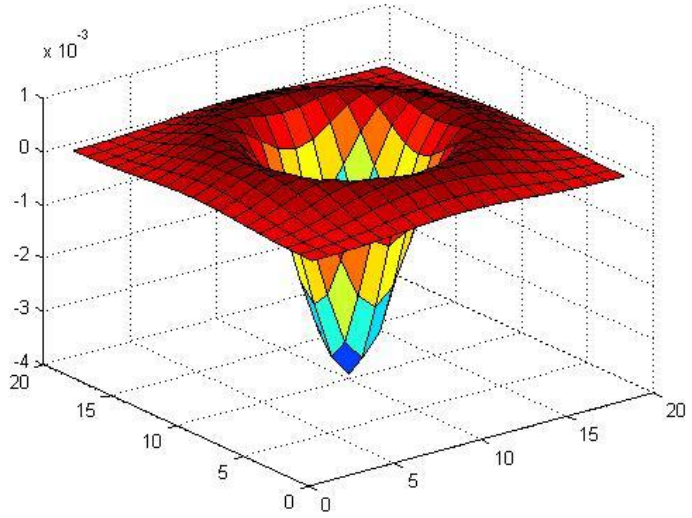


Difference of Gaussian (DoG)

$$g(\sigma_1) * I - g(\sigma_2) * I = [g(\sigma_1) - g(\sigma_2)] * I$$

- Looks a lot like LoG!
- (Not actually the same, but similar)





Difference of Gaussian (DoG)



$\sigma = 1$

-



original

=



result (contrast enhanced)

Difference of Gaussian (DoG)



$\sigma = 2$

-



$\sigma = 1$

=



result (contrast enhanced)

Difference of Gaussian (DoG)



$\sigma = 3$

-



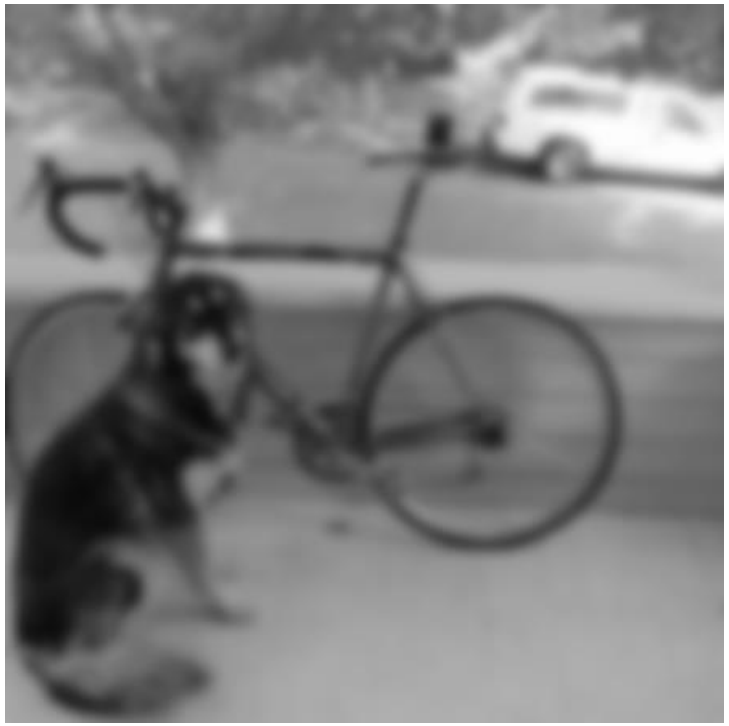
$\sigma = 2$

=



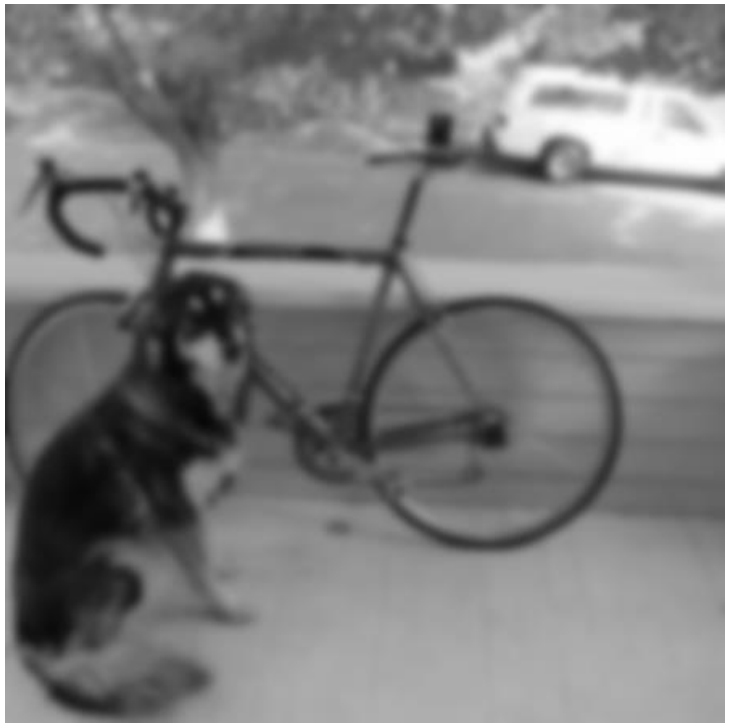
result (contrast enhanced)

Difference of Gaussian (DoG)



$\sigma = 4$

-



$\sigma = 3$

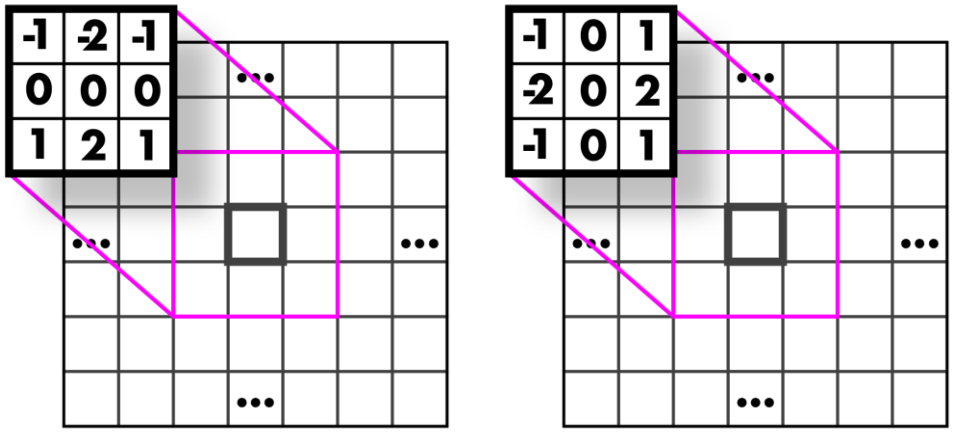
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result (contrast enhanced)

Edge detection

- Back to gradients: Use magnitude of gradients
- Sobel kernels



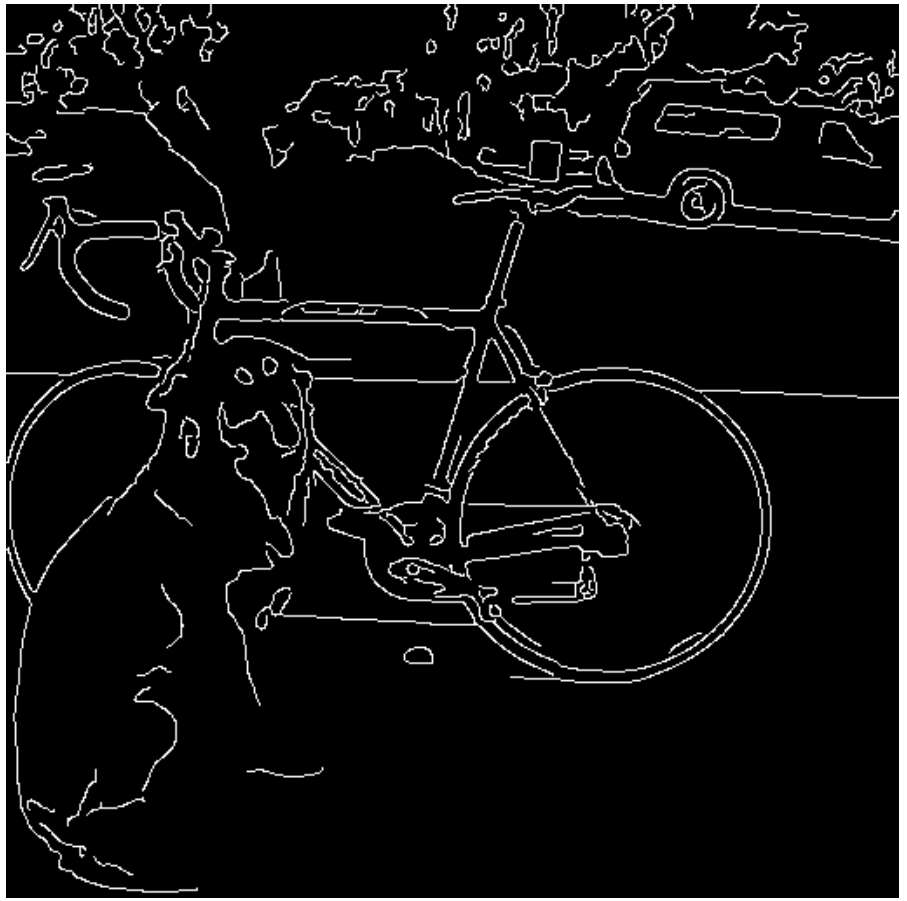




Edge detection

- What we want: Line drawing!





Canny edge detector

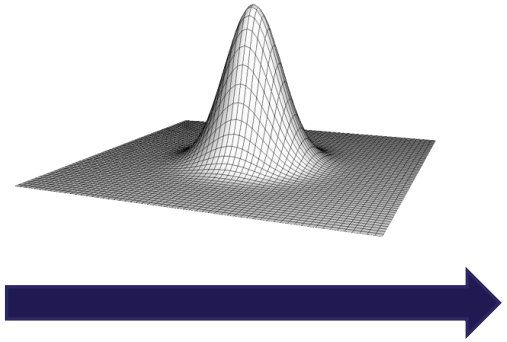
Algorithm:

1. Smooth image (Gaussian filter)
2. Calculate gradient direction and magnitude
3. Non-maximum suppression perpendicular to edge
4. Threshold into strong, weak, no edge
5. Connect components

Canny edge detector

- Smoothing – Gaussian filter

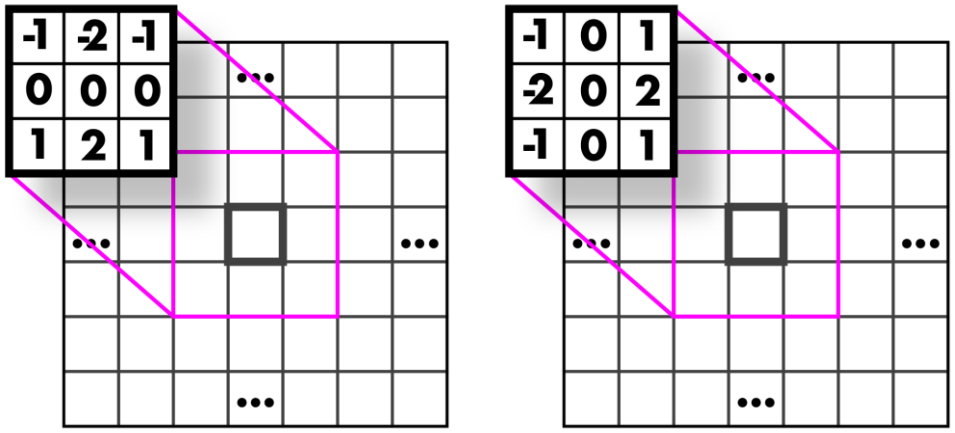






Canny edge detector

- Gradient magnitude and direction
  - Sobel filter



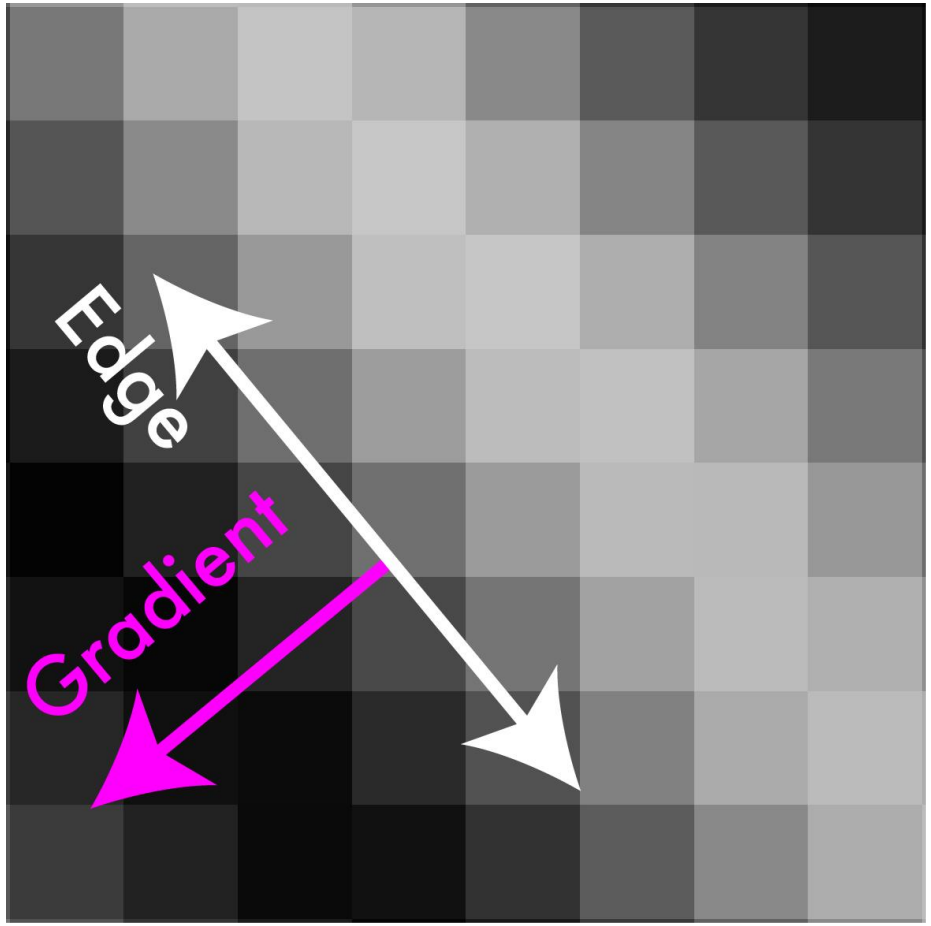


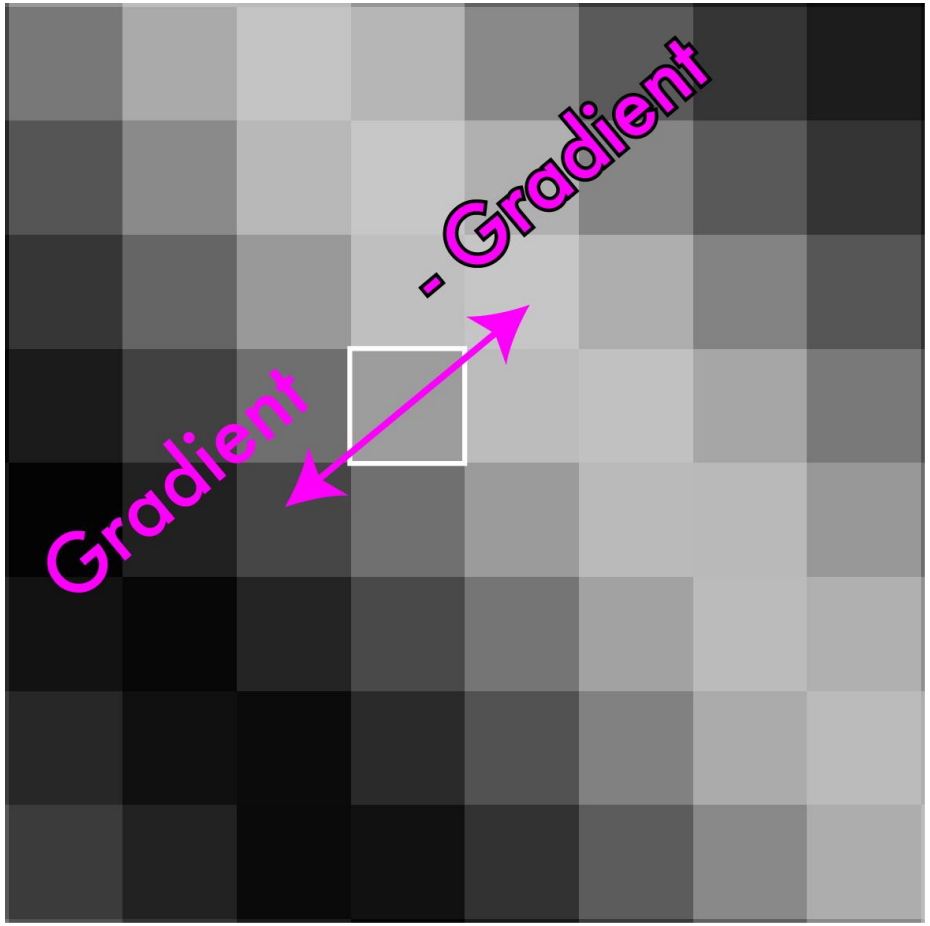
Canny edge detector

- Non-maximum suppression
  - Goal: Single pixel edges, not thick blurry lines
  - Need to check nearby pixels and keep only highest respons

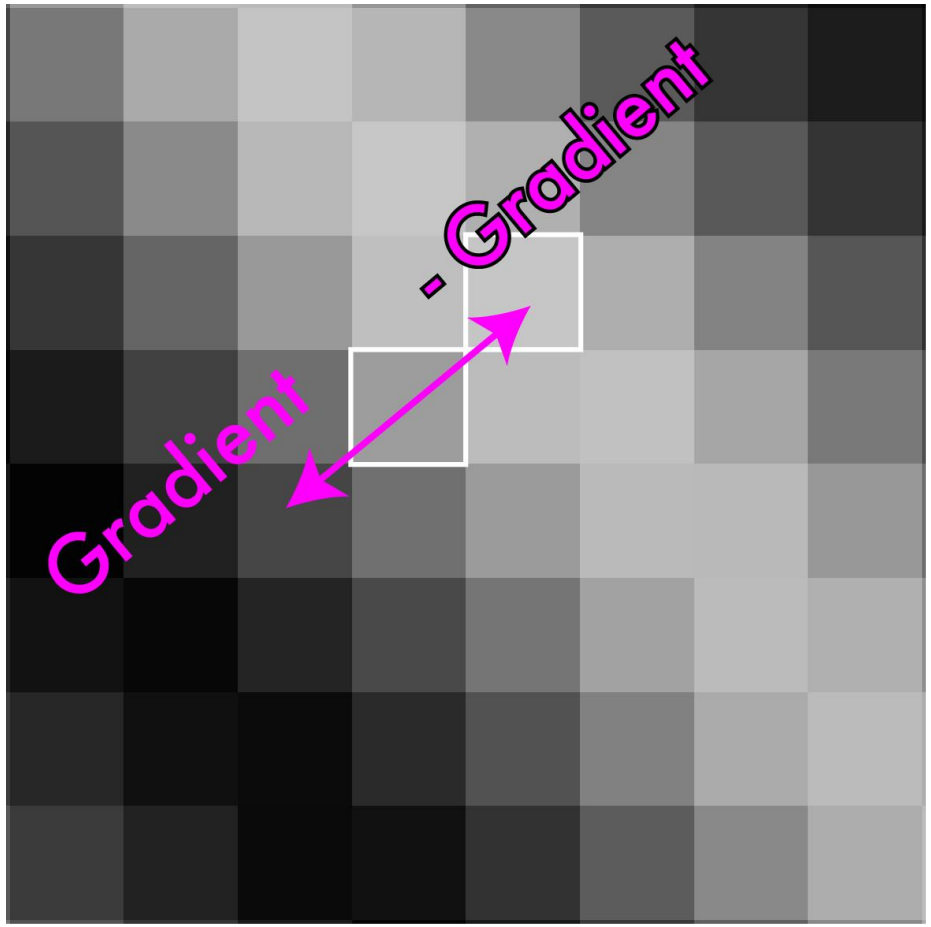


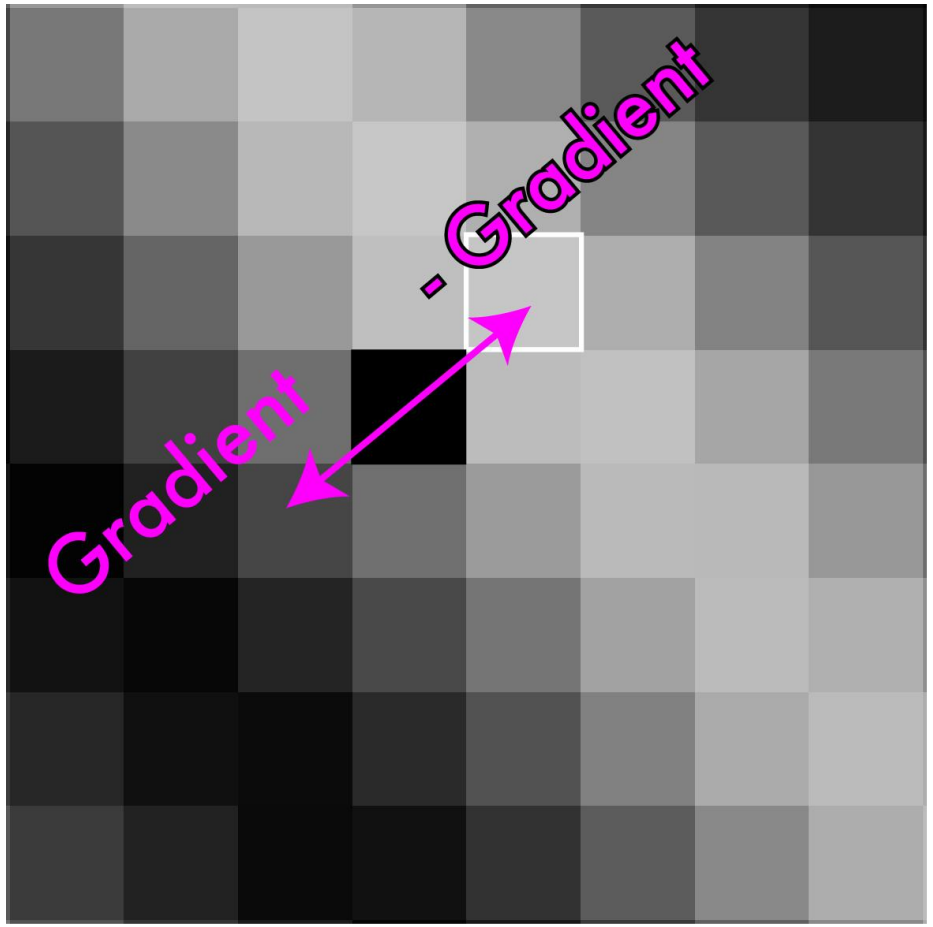
Canny edge detector



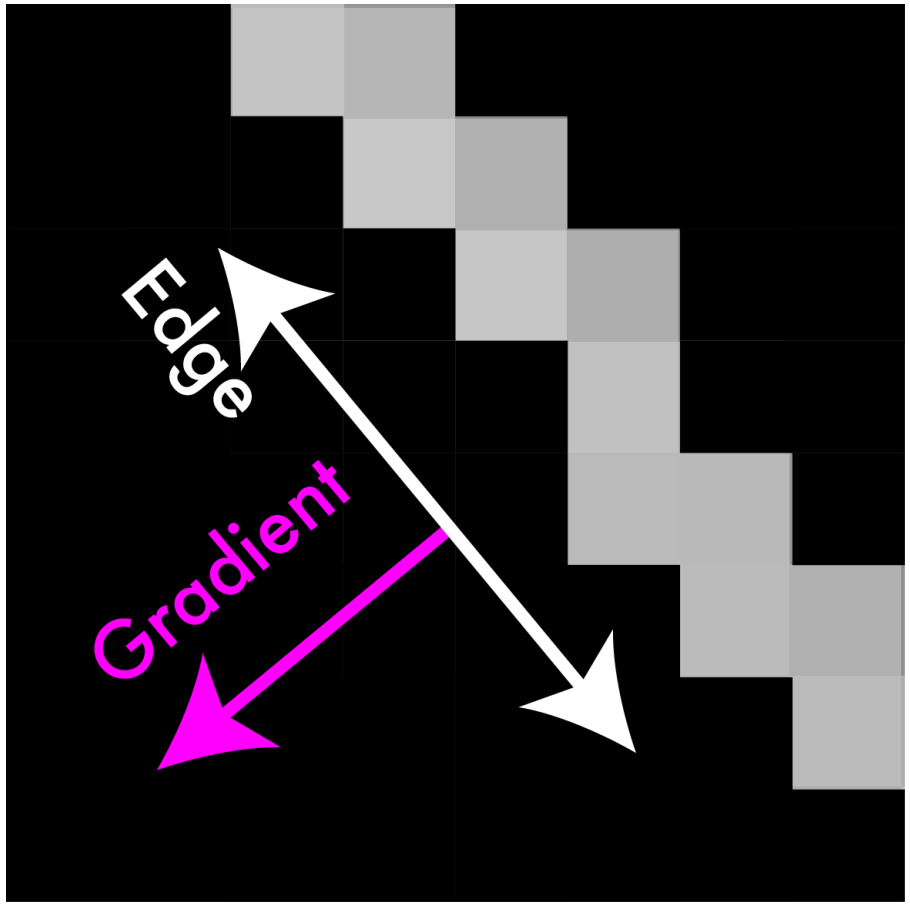


Canny edge detector

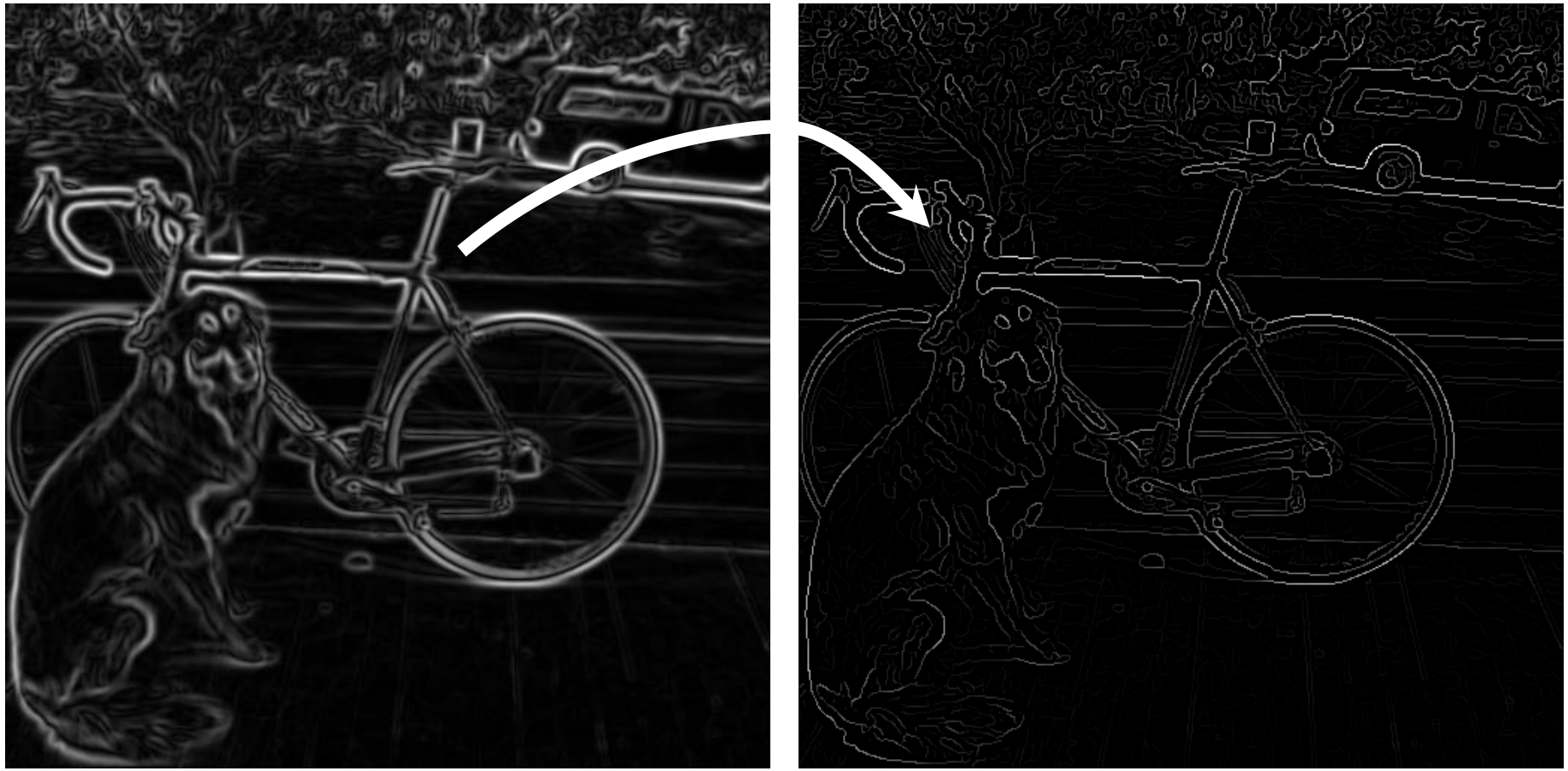




Canny edge detector



Canny edge detector



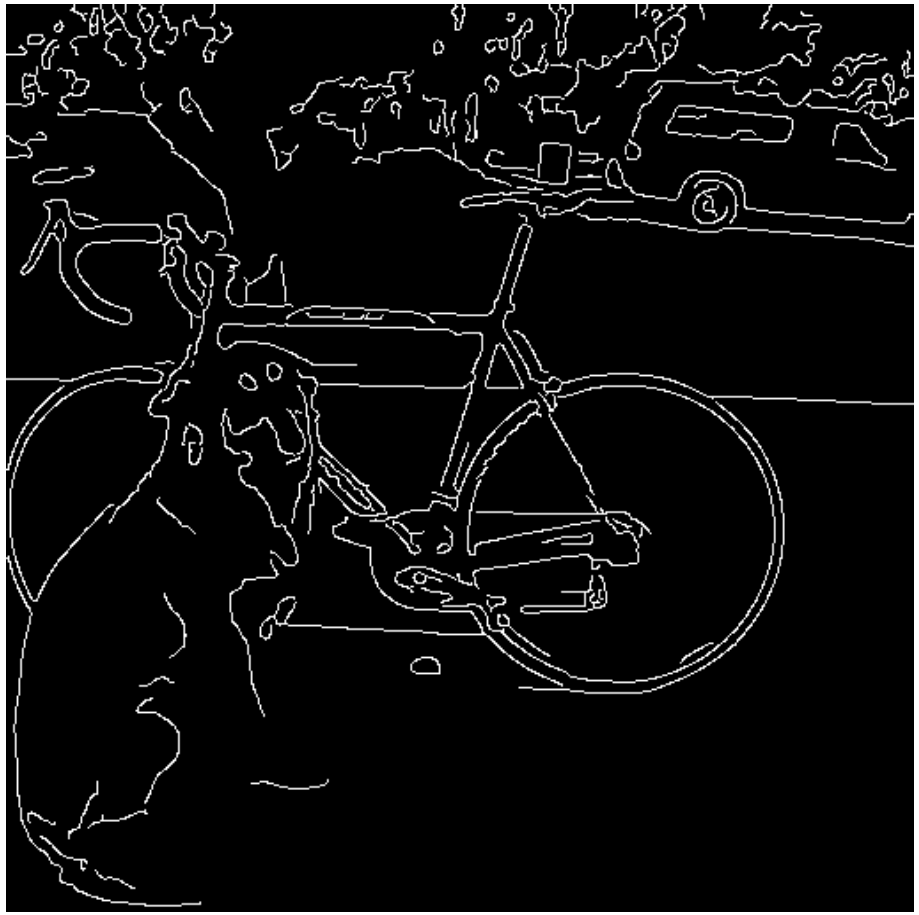
Canny edge detector

- Still some noise
- Only want strong edges
- 2 thresholds – 3 cases
  - PixVal > T_{high}: strong edge
  - T_{high} > PixVal > T_{low}: weak edge
  - PixVal < T_{low}: no edge



Canny edge detector

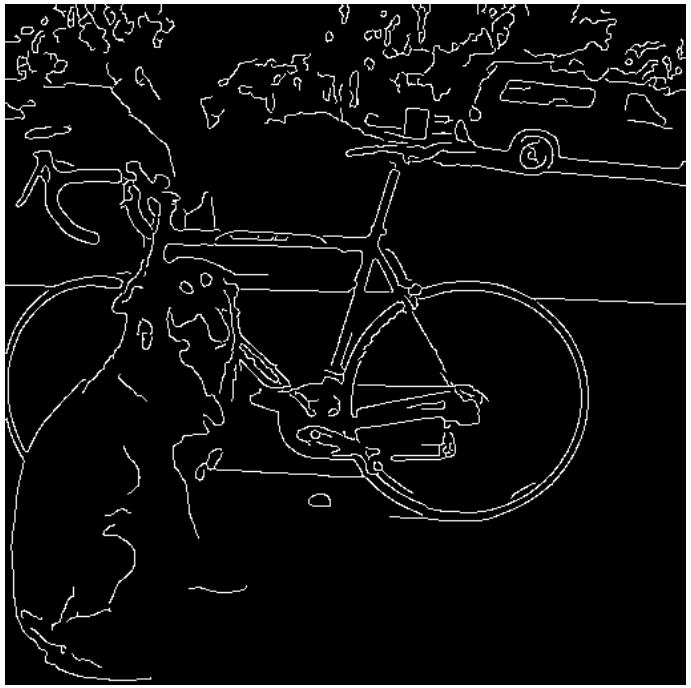
- Strong edge are edges!
- Weak edges are edges if they connect to strong edges
  - Look in some neighborhood (usually 8 closest)
  - Hysteresis



Canny edge detector

- Tunable parameters: Sigma (Gaussian), thresholds T_{high} and T_{low}
- Demo: http://bigwww.epfl.ch/demo/ip/demos/edgeDetector/





Summary

- Correlation is the basic procedure behind almost all IP / CV
- Edge detection: Based on derivatives (1st or 2nd)
  - Sobel kernels
  - Canny edge detection algorithm
- Edge detection/gradients is the basis for many feature detection methods, to be continued in next lecture

Exercises on Moodle:

- Getting started with OpenCV and play with edge detection